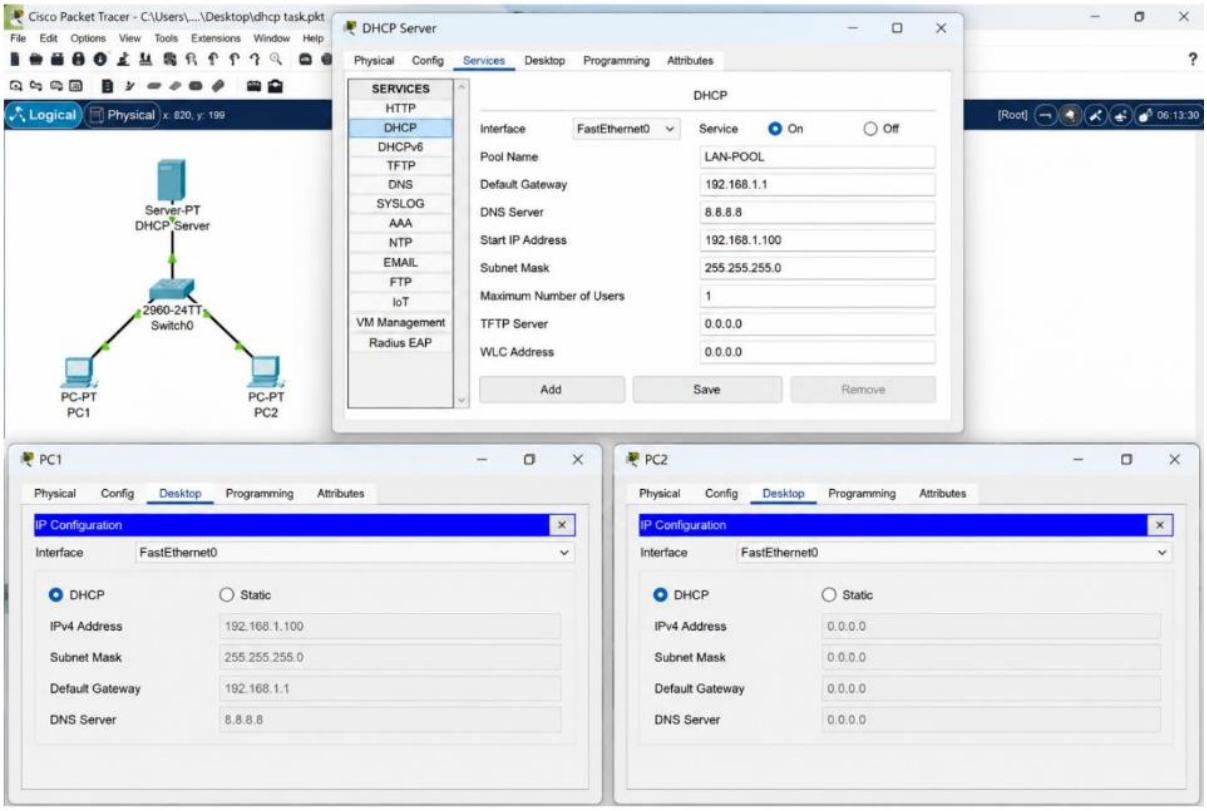
1.

Set up Windows Defender Firewall on your Windows laptop or PC and create a new inbound rule to block all incoming traffic on port 8080. Take a screenshot of the rule you created.

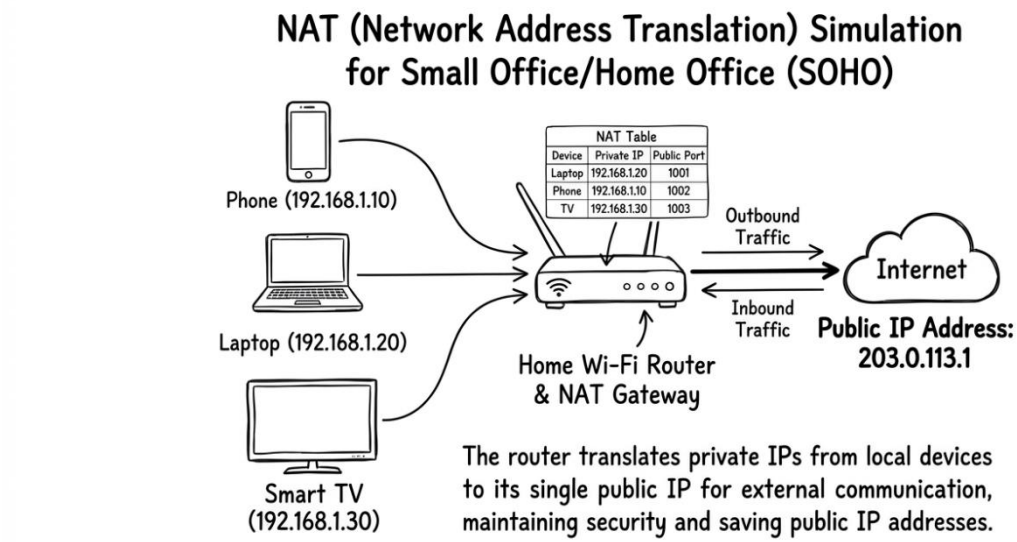
ANS:-



2.

Simulate NAT (Network Address Translation) by drawing a simple diagram showing how your home WiFi router assigns private IP addresses to your phone, laptop, and smart TV, and how it translates them to a single public IP for internet access. Hint: Use any drawing tool or even hand-draw and click a photo.

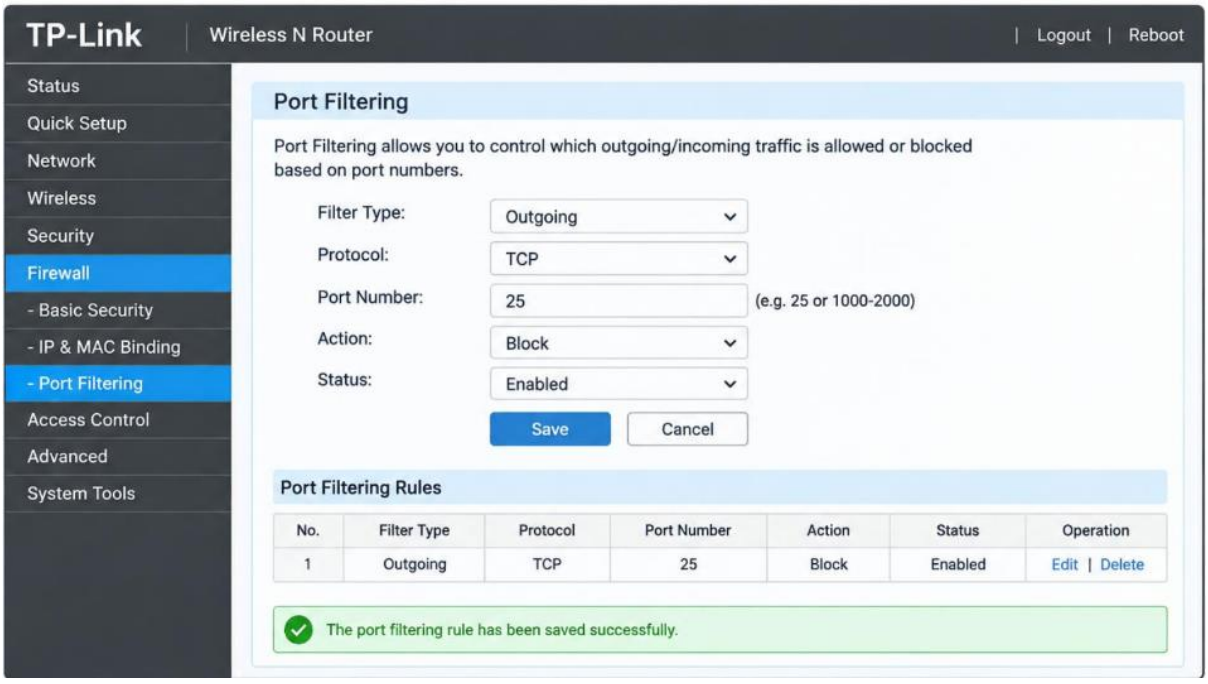
ANS:-



3.

Configure port filtering using your WiFi router's admin panel (or use a router simulator online if you don't have access). Block outgoing traffic on port 25 (SMTP) and explain in 2 sentences how this could help prevent spam from your network.

ANS:-



Explanation: Blocking outgoing port 25 prevents compromised local devices or malware from directly connecting to external mail servers to transmit unsolicited bulk emails (spam). This protects your network's public IP address from being blacklisted by email service providers.

4.

Given a sample packet filtering rule set (provided by your trainer), identify which of these three sample packets would be allowed or blocked: 1) HTTP request from 192.168.1.5 to 172.217.166.78:80, 2) SSH request from 10.0.0.10 to 34.201.23.1:22, 3) DNS query from 192.168.1.7 to 8.8.8.8:53. Hint: Focus on source/destination IPs and ports in the rule set

ANS:-

Assuming a standard secure firewall rule set (which allows outbound DNS and HTTP/HTTPS traffic while blocking unapproved outgoing ports like SSH):

1. **HTTP request from 192.168.1.5 to 172.217.166.78:80**
    - **Result: Allowed**
    - **Reason:** Outbound web traffic on port 80 (HTTP) is permitted for browsing and app connectivity.
2. **SSH request from 10.0.0.10 to 34.201.23.1:22**
    - **Result: Blocked**
    - **Reason:** Outbound remote management ports like SSH (port 22) are typically restricted or blocked by default firewalls to prevent unauthorized administrative access.
3. **DNS query from 192.168.1.7 to 8.8.8.8:53**
    - **Result: Allowed**
    - **Reason:** Outbound DNS traffic on port 53 is required for domain name resolution so devices can look up website addresses.

5.

Research and list 3 ways Instagram or Zomato might use firewalls and packet filtering to protect their servers from attacks or misuse.

ANS:-

Large-scale platforms like Instagram and Zomato utilize advanced firewalls, Web Application Filters (WAFs), and packet inspection systems in the following ways:

1. **DDoS Mitigation & Rate Limiting (Layer 4/7 Filtering):** Firewalls inspect incoming traffic headers and volume, automatically dropping packets or rate-limiting IP addresses that flood servers with malicious requests during Distributed Denial of Service (DDoS) attacks.
2. **Database & Microservices Network Segmentation:** Internal firewalls restrict direct public access to backend database clusters (such as user profile databases or transaction ledgers), ensuring that database ports are only accessible internally via authorized application microservices.
3. **Geographic and Threat Intelligence Blocking:** Firewalls analyze packet source locations and threat intelligence feeds to block traffic originating from known malicious IP ranges, botnets, or unauthorized geographic regions where the service does not operate.